



INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions

AP[®] Chemistry

COURSE AND EXAM DESCRIPTION

Effective
Fall 2024

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008	2 He 4.00											13 B 10.81	14 C 12.01	15 N 14.01	16 O 16.00	17 F 19.00	18 Ne 20.18
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.30	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc 101.07	44 Ru 102.91	45 Rh 106.42	46 Pd 107.87	47 Ag 112.41	48 Cd 114.82	49 In 118.71	50 Sn 121.76	51 Sb 127.60	52 Te 126.90	53 I 131.29	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57-71 *	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po	85 At	86 Rn
87 Fr	88 Ra	89-103 †	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

*Lanthanoids

57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm	62 Sm 150.36	63 Eu 151.97	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97
89 Ac	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

†Actinoids

AP® CHEMISTRY EQUATIONS AND CONSTANTS, EFFECTIVE 2025

UNIT SYMBOLS	
gram,	g
mole,	mol
liter,	L
meter,	m
second,	s
hertz,	Hz
atmosphere,	atm
millimeter of mercury,	mm Hg
degree Celsius,	°C
kelvin,	K
joule,	J
volt,	V
coulomb,	C
ampere,	A

UNIT CONVERSIONS
1 hertz = 1 s ⁻¹
1 atm = 760 mm Hg = 760 torr
K = °C + 273.15
1 volt = $\frac{1 \text{ joule}}{1 \text{ coulomb}}$
1 ampere = $\frac{1 \text{ coulomb}}{1 \text{ second}}$

METRIC PREFIXES		
Factor	Prefix	Symbol
10 ⁹	giga	G
10 ⁶	mega	M
10 ³	kilo	k
10 ⁻²	centi	c
10 ⁻³	milli	m
10 ⁻⁶	micro	μ
10 ⁻⁹	nano	n
10 ⁻¹²	pico	p

ATOMIC STRUCTURE

$$E = h\nu$$

$$c = \lambda\nu$$

$$F_{\text{coulombic}} \propto \frac{q_1 q_2}{r^2}$$

E = energy
 ν = frequency
 λ = wavelength
 F = force
 q = charge
 r = separation

Planck's constant, $h = 6.626 \times 10^{-34}$ J s
 Speed of light, $c = 2.998 \times 10^8$ m s⁻¹
 Avogadro's number = 6.022×10^{23} mol⁻¹

GASES, LIQUIDS, AND SOLUTIONS

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$PV = nRT$$

$$P_A = P_{\text{total}} \times X_A, \text{ where } X_A = \frac{\text{moles A}}{\text{total moles}}$$

$$P_{\text{total}} = P_A + P_B + P_C + \dots$$

$$n = \frac{m}{M}$$

$$D = \frac{m}{V}$$

$$KE = \frac{1}{2}mv^2$$

$$M = \frac{n_{\text{solute}}}{L_{\text{solution}}}$$

$$A = \epsilon bc$$

P = pressure
 V = volume
 T = temperature
 n = number of moles
 X = mole fraction
 m = mass
 M = molar mass
 D = density
 KE = kinetic energy
 ν = velocity
 M = molarity
 A = absorbance
 ϵ = molar absorptivity
 b = path length
 c = concentration

Gas constant, $R = 8.314$ J mol⁻¹ K⁻¹
 $= 0.08206$ L atm K⁻¹ mol⁻¹
 STP = 273.15 K and 1.0 atm
 Ideal gas at STP = 22.4 L mol⁻¹

KINETICS

$$[A]_t - [A]_0 = -kt$$

$$\ln[A]_t - \ln[A]_0 = -kt$$

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$

$$t_{1/2} = \frac{0.693}{k}$$

k = rate constant
 t = time
 $t_{1/2}$ = half-life

EQUILIBRIUM

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}, \text{ where } a A + b B \rightleftharpoons c C + d D$$

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

$$K_w = [H_3O^+][OH^-] = 1.0 \times 10^{-14} \text{ at } 25^\circ\text{C}$$

$$pK_w = 14 = \text{pH} + \text{pOH at } 25^\circ\text{C}$$

$$\text{pH} = -\log[H_3O^+], \quad \text{pOH} = -\log[OH^-]$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}, \quad K_b = \frac{[OH^-][HB^+]}{[B]}$$

$$pK_a = -\log K_a, \quad pK_b = -\log K_b$$

$$K_w = K_a \times K_b, \quad pK_w = pK_a + pK_b$$

$$\text{pH} = pK_a + \log \frac{[A^-]}{[HA]}$$

Equilibrium Constants

K_c (molar concentrations)
 K_p (gas pressures)
 K_w (water)
 K_a (acid)
 K_b (base)

THERMODYNAMICS/ELECTROCHEMISTRY

$$q = mc\Delta T$$

$$\Delta H^\circ_{\text{reaction}} = \sum \Delta H^\circ_{f \text{ products}} - \sum \Delta H^\circ_{f \text{ reactants}}$$

$$\Delta S^\circ_{\text{reaction}} = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$$

$$\Delta G^\circ_{\text{reaction}} = \sum \Delta G^\circ_{f \text{ products}} - \sum \Delta G^\circ_{f \text{ reactants}}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= -RT \ln K$$

$$= -nFE^\circ$$

$$I = \frac{q}{t}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q$$

q = heat
 m = mass
 c = specific heat capacity
 T = temperature
 S° = standard entropy
 H° = standard enthalpy
 G° = standard Gibbs free energy
 R = gas constant
 K = equilibrium constant
 n = number of moles of electrons
 E° = standard potential
 I = current (amperes)
 q = charge (coulombs)
 t = time (seconds)
 Q = reaction quotient

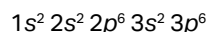
Faraday's constant, $F = 96,485 \text{ coulombs / 1 mol } e^-$

Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and AP Chemistry Exam and serve as examples of the types of questions that appear on the exam. After the sample questions is a table that shows which skill, learning objective(s), and unit each question relates to. The table also provides the answers to the multiple-choice questions.

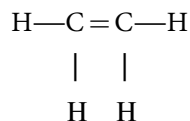
Section I: Multiple-Choice

The following are examples of the kinds of multiple-choice questions found on the exam.



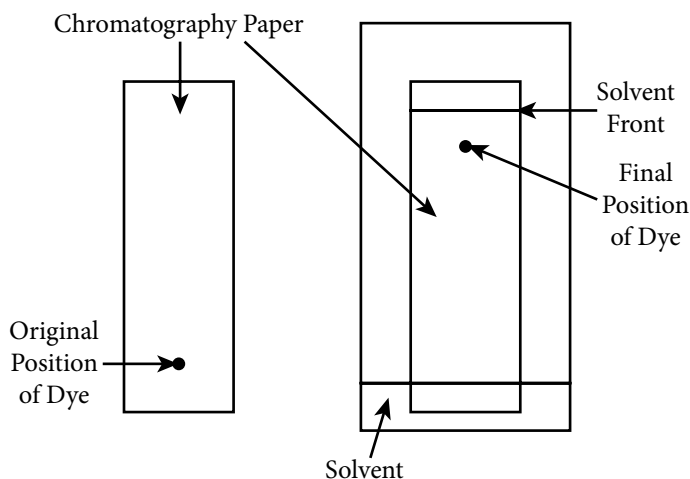
1. Which of the following species has the electron configuration shown above?

- (A) O
- (B) Ne
- (C) K^+
- (D) Cl^+

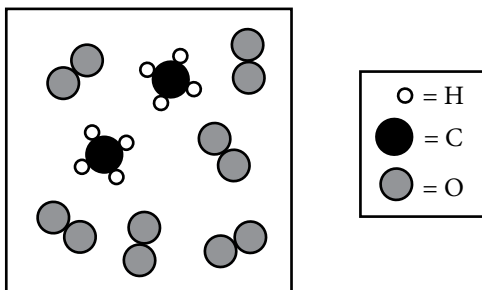


2. A Lewis diagram for the molecule C_2H_4 is shown above. In the actual C_2H_4 molecule, the H-C-H bond angles are closest to

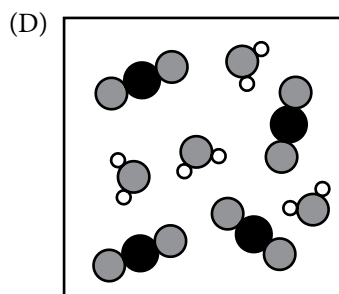
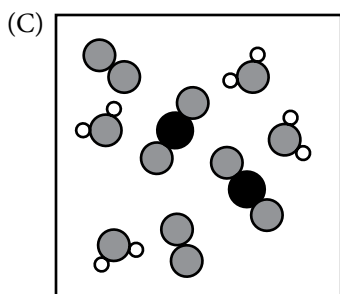
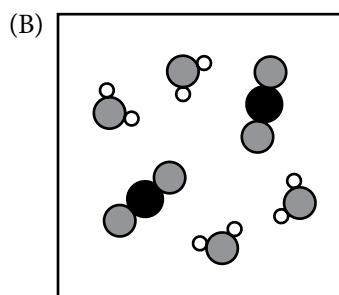
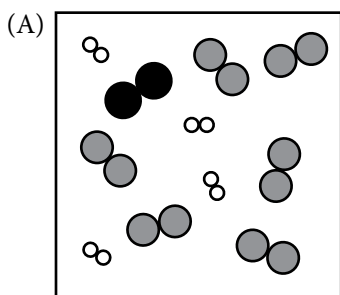
- (A) 90°
- (B) 109.5°
- (C) 120°
- (D) 180°



3. Based on the results of the paper chromatography experiment shown above, which of the following can be concluded about the dye?
- (A) It has a small molar mass.
- (B) It has weak intermolecular forces.
- (C) It has a weaker attraction for the stationary phase than it has for the mobile phase.
- (D) It has a stronger attraction for the stationary phase than it has for the mobile phase.



4. The reactants represented above are placed in a vessel and a reaction occurs. Which of the following best represents the contents of the vessel after the reaction has proceeded as completely as possible?

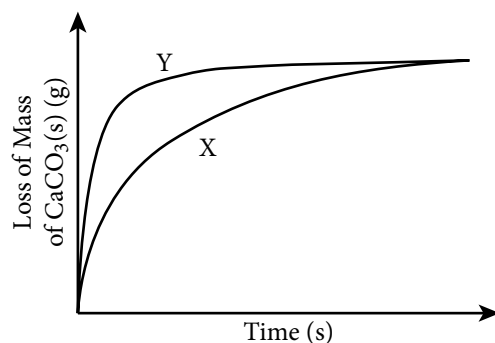


Initial buret reading	2.17 mL
Final buret reading	22.17 mL

5. An antacid tablet containing $\text{Mg}(\text{OH})_2(s)$ (molar mass 58.3 g/mol) is titrated with a 0.100 M solution of $\text{HNO}_3(aq)$. The end point is determined by using an indicator. Based on the data in the table above, what was the mass of the $\text{Mg}(\text{OH})_2(s)$ in the antacid tablet?
- (A) 0.0583 g
 (B) 0.583 g
 (C) 5.83 g
 (D) 58.3 g



The reaction between $\text{HCl}(aq)$ and $\text{CaCO}_3(s)$ is represented by the equation above. Two separate trials were carried out using $\text{CaCO}_3(s)$ samples of the same mass, but one sample was a single piece of $\text{CaCO}_3(s)$, and one sample was composed of small pieces of $\text{CaCO}_3(s)$. The loss of mass of $\text{CaCO}_3(s)$ as a function of time for both trials is shown in the graph below.



6. Which of the curves, X or Y, represents the reaction with small pieces of $\text{CaCO}_3(s)$, and why?
- (A) Curve X, because it shows that the reaction proceeded at a uniform rate.
 (B) Curve X, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $\text{CaCO}_3(s)$.
 (C) Curve Y, because it shows that the reaction proceeded at a nonuniform rate.
 (D) Curve Y, because it takes a shorter time for the reaction to go to completion due to the larger surface area of $\text{CaCO}_3(s)$.

Cup	Material of Spoon	Initial Temperature of Spoon (°C)	Mass of Spoon (g)	Specific Heat Capacity (J/(g · °C))
A	Aluminum	20	10.0	0.90
B	Ceramic	20	10.0	0.80
C	Steel	20	20.0	0.45
D	Silver	20	40.0	0.23

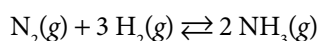
7. Four identical 50 mL cups of coffee, originally at 95°C, were stirred with four different spoons, as listed in the table above. In which cup will the temperature of the coffee be highest at thermal equilibrium? (Assume that the heat lost to the surroundings is negligible.)

(A) Cup A
(B) Cup B
(C) Cup C
(D) Cup D



8. Equal volumes of 0.1 M $\text{AgNO}_3(aq)$ and 0.4 M $\text{NH}_3(aq)$ are mixed and allowed to reach the equilibrium represented above. Which of the following correctly lists the equilibrium concentrations of the $\text{Ag}^+(aq)$, $\text{NH}_3(aq)$, and $[\text{Ag}(\text{NH}_3)_2]^+(aq)$ in order from least to greatest?

(A) $[\text{Ag}^+]_{eq} < [\text{NH}_3]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq}$
(B) $[\text{NH}_3]_{eq} < [\text{Ag}^+]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq}$
(C) $[\text{Ag}(\text{NH}_3)_2]^+_{eq} < [\text{Ag}^+]_{eq} < [\text{NH}_3]_{eq}$
(D) $[\text{Ag}^+]_{eq} < [[\text{Ag}(\text{NH}_3)_2]^+]_{eq} < [\text{NH}_3]_{eq}$

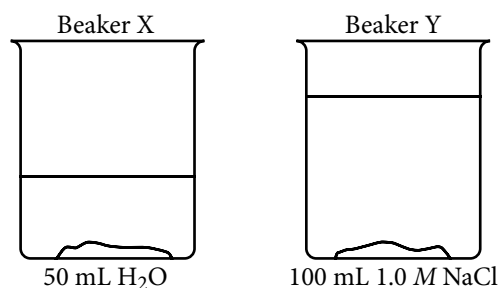


A 2.0 L reaction vessel contains an equilibrium mixture of the system represented above. The partial pressures of the components of the mixture at equilibrium are listed in the table below.

P_{N_2}	P_{H_2}	P_{NH_3}
0.30 atm	0.50 atm	0.20 atm

9. Which of the following is the best estimate of the total pressure of the system when the volume of the container is reduced to 1.0 L at a constant temperature?

(A) $P_{total} = 1.00 \text{ atm}$
(B) $1.00 \text{ atm} < P_{total} < 2.00 \text{ atm}$
(C) $P_{total} = 2.00 \text{ atm}$
(D) $P_{total} > 2.00 \text{ atm}$



10. Beaker X contains 50 mL of distilled water and beaker Y contains 100 mL of 1.0 M NaCl. Solid AgCl is added to each of the beakers. After thoroughly stirring the contents of the beakers, some solid AgCl remains at the bottom of each beaker, as shown above. Which of the following is true?
- (A) $[\text{Ag}^+]$ is zero in both beakers.
 (B) $[\text{Ag}^+]$ is the same, but not zero, in both beakers.
 (C) $[\text{Ag}^+]$ is greater in beaker X.
 (D) $[\text{Ag}^+]$ is greater in beaker Y.
11. The value of K_w for water at 0°C is 1×10^{-15} . What is the pOH of water at 0°C ?
- (A) 6.5
 (B) 7.0
 (C) 7.5
 (D) 8.0

Half-Reaction	$E^\circ(\text{V})$
$\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s)$	0.80
$\text{Cr}^{3+}(aq) + 3 e^- \rightarrow \text{Cr}(s)$	-0.41

12. Based on the standard reduction potentials in the table above, what is the value of E° for a standard galvanic cell made with Ag/Ag⁺ and Cr/Cr³⁺ half-cells?
- (A) 0.39 V
 (B) 1.21 V
 (C) 1.99 V
 (D) 2.81 V

Questions 13–15 refer to the information below.

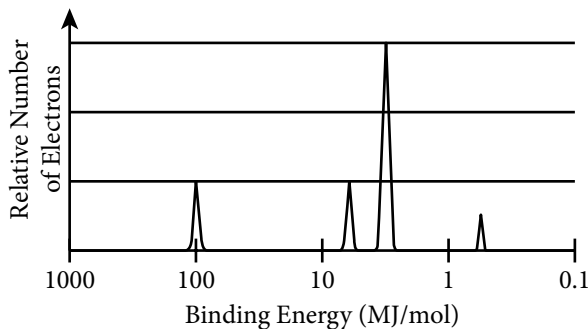
A 0.50 g sample of Mg(s) was placed in a solution of HCl(aq), where it reacted completely.

13. Which of the following equations best represents the reacting species in the reaction described above?
- (A) $\text{Mg}(s) + 2 \text{H}^+(aq) \rightarrow \text{Mg}^{2+}(aq) + \text{H}_2(g)$
 (B) $\text{Mg}(s) + \text{HCl}(aq) \rightarrow \text{MgCl}_2(aq) + \text{H}_2(g)$
 (C) $\text{Mg}(s) + 2 \text{HCl}(aq) \rightarrow \text{MgCl}_2(s) + 2 \text{H}^+(aq)$
 (D) $\text{Mg}(s) + 2 \text{HCl}(aq) \rightarrow \text{Mg}^{2+}(aq) + \text{Cl}_2(g) + \text{H}_2(g)$

14. In another experiment, a small piece of $\text{Mg}(s)$ is weighed, then placed in a flask containing excess $1\text{ M HCl}(aq)$. The student wants to determine the number of moles of gas produced. Which of the following is the best way to conduct the experiment for accurate data collection?
- (A) Conducting the experiment at different temperatures to see which generates the most gas
- (B) Completing the entire reaction in a large Erlenmeyer flask of known volume to measure the volume of the gas collected
- (C) Collecting the gas in a eudiometer tube and measuring the volume of the gas collected
- (D) Conducting the reaction in a graduated cylinder and measuring the volume of the gas collected
15. In a third experiment, 0.10 g samples of $\text{Mg}(s)$ are placed in excess $\text{HCl}(aq)$ of various concentrations: 0.050 M , 0.10 M , 0.25 M , and 0.50 M . The reactions are run in successive order from 0.050 M to 0.50 M , and the time required for each reaction to go to completion is recorded. As the concentration of $\text{HCl}(aq)$ increases from 0.050 M to 0.50 M , which of the following is the expected result?
- (A) The reaction time increases, and the rate of the reaction decreases.
- (B) The reaction time decreases, and the rate of the reaction increases.
- (C) Both the reaction time and the rate of the reaction increase.
- (D) Both the reaction time and the rate of the reaction decrease.

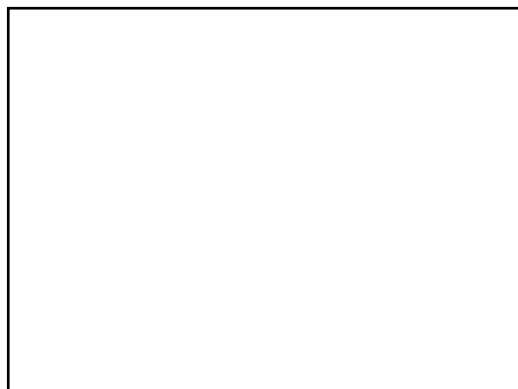
Section II: Free-Response

The following are examples of the kinds of free-response questions found on the exam. Note that on the actual AP Exam, there will be three long free-response questions and four short free-response questions.



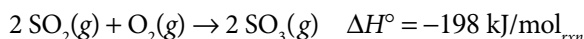
1. The complete photoelectron spectrum of an unknown element is given above.
- (a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb's law.
- (b) Based on the spectrum, write the complete electron configuration of the element.
- (c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has.

2. The following questions relate to sulfur and some of its compounds.
- (a) Write the balanced equation for the combustion of $\text{S}_8(\text{s})$ to form $\text{SO}_2(\text{g})$.
 - (b) Calculate the volume of $\text{O}_2(\text{g})$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $\text{S}_8(\text{s})$.
 - (c) A student claims that the combustion of S_8 is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction.
 - (d) In the box below, draw a Lewis electron-dot diagram for one valid resonance structure of SO_2 .

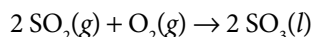


- (e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2 ?

SO_2 can be oxidized to form SO_3 according to the following equation.



- (f) Is the value of ΔS° for the reaction represented above positive or negative? Justify your answer.



- (g) Is the magnitude of ΔH° for the reaction to form $\text{SO}_3(\text{l})$, represented above, greater than, less than, or equal to the magnitude of ΔH° for the reaction to form $\text{SO}_3(\text{g})$? Justify your answer.
- (h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Essential Knowledge
1	C	1.A	1.5.A	1.5.A.3
2	C	6.C	2.7.A	2.7.A.2
3	C	2.D	3.9.A	3.9.A.1
4	C	5.C	4.5.A	4.5.A.1
5	A	5.F	4.6.A	4.6.A.1
6	D	6.F	5.1.A	5.1.A.3
7	B	5.F	6.4.A	6.4.A.1
8	D	5.F	7.7.A	7.7.A.1
9	B	5.F	7.9.A	7.9.A.1
10	C	5.F	7.12.A	7.12.A.1
11	C	5.F	8.1.A	8.1.A.4
12	B	5.F	9.9.A	9.9.A.2
13	A	5.E	4.2.A	4.2.A.3
14	C	2.C	3.4.A	3.4.A.1
15	B	2.B	5.2.A	5.2.A.1

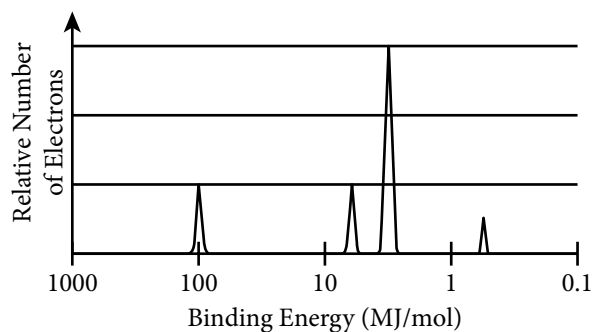
Free-Response Question	Question Type	Skill	Learning Objective
1	Short	3.A, 3.B, 6.D	1.6.A
2	Long	1.A, 3.B, 4.C, 5.C, 5.E, 5.F, 6.D	2.5.A, 2.7.A, 4.2.A, 4.5.A, 4.7.A, 6.9.B, 9.1.A, 9.3.A

The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the [AP Chemistry Exam Page](#) on AP Central.



Scoring Guidelines

Question 1: Short-Answer



1. The complete photoelectron spectrum of an unknown element is given above.
 - (a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb's law.
 - (b) Based on the spectrum, write the complete electron configuration of the element.
 - (c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has.

Scoring Guidelines for Question 1: Short-Answer

4 points

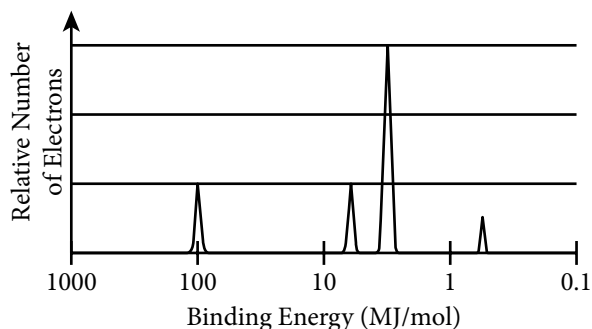
Learning Objectives: 1.6.A

- (a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb's law.

1 point

6.D

1.6.A



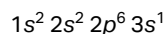
See the student's drawing.

One point for the X located above the leftmost peak (at 100 MJ/mol) in the spectrum with the justification that the electrons closest to the nucleus have the greatest binding energy because the strength of attraction between the charges (electron and nucleus) is greatest when the distance between them (r) is the least.

- (b) Based on the spectrum, write the complete electron configuration of the element.

1 point

One point for the correct configuration.



3.B

1.6.A

- (c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has.

1 point

3.A

See the student's drawing.

1.6.A

One point if the peak is located just to the left of the rightmost peak in the spectrum.

One point if the height of the peak is twice the height of the rightmost peak in the spectrum.

1 point

3.A

1.6.A

Total for part (c)

2 points

Total for question 1

4 points

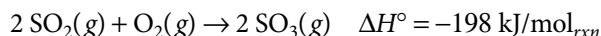
Question 2: Long-Answer

2. The following questions relate to sulfur and some of its compounds.
- (a) Write the balanced equation for the combustion of $\text{S}_8(\text{s})$ to form $\text{SO}_2(\text{g})$.
 - (b) Calculate the volume of $\text{O}_2(\text{g})$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $\text{S}_8(\text{s})$.
 - (c) A student claims that the combustion of S_8 is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction.
 - (d) In the box below, draw a Lewis electron-dot diagram for one valid resonance structure of SO_2 .

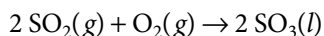


- (e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2 ?

SO_2 can be oxidized to form SO_3 according to the following equation.



- (f) Is the value of ΔS° for the reaction represented above positive or negative? Justify your answer.



- (g) Is the magnitude of ΔH° for the reaction to form $\text{SO}_3(\text{l})$, represented above, greater than, less than, or equal to the magnitude of ΔH° for the reaction to form $\text{SO}_3(\text{g})$? Justify your answer.
- (h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

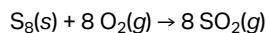
Scoring Guidelines for Question 2: Long-Answer

10 points

Learning Objectives: 2.5.A 2.7.A 4.2.A 4.5.A 4.7.A 6.9.B 9.1.A 9.3.A

- (a) Write the balanced equation for the combustion of $S_8(s)$ to form $SO_2(g)$. **1 point**

One point for the balanced equation.



5.E

4.2.A

- (b) Calculate the volume of $O_2(g)$, measured at 1.00 atm and 298 K, that is required to completely combust a 500.0 g sample of pure $S_8(s)$. **1 point**

One point for calculating the moles of O_2 .

$$500.0 \text{ g } S_8 \times \frac{1 \text{ mol } S_8}{256.5 \text{ g } S_8} \times \frac{8 \text{ mol } O_2}{1 \text{ mol } S_8} = 15.60 \text{ mol } O_2$$

5.F

4.5.A

One point for calculating the volume of O_2 .

$$V = \frac{nRT}{P} = \frac{(15.60 \text{ mol})(0.08206 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \cdot \text{K}^{-1})(298 \text{ K})}{1.00 \text{ atm}} = 381 \text{ L}$$

1 point

5.F

4.5.A

Total for part (b) 2 points

- (c) A student claims that the combustion of $S_8(s)$ is an oxidation-reduction reaction. Justify the claim by identifying the oxidation numbers of sulfur and oxygen both before and after the reaction. **1 point**

One point for all four correct oxidation numbers.

Oxidation numbers before the reaction: S = 0, O = 0

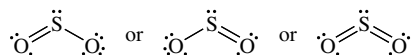
Oxidation numbers after the reaction: S = +4, O = -2

1.A

4.7.A

- (d) In the box below, draw a complete Lewis electron-dot diagram for one valid resonance structure of $SO_2(g)$. **1 point**

Any one of the three following diagrams is acceptable.



One point for the correct number of electrons.

3.B

2.5.A

One point for a valid Lewis diagram.

1 point

3.B

2.5.A

Total for part (d) 2 points

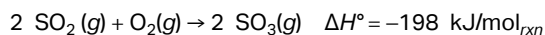
- (e) Based on the diagram you drew in part (d), what is the approximate oxygen-sulfur-oxygen bond angle in SO_2 ? **1 point**

One point for an angle that is consistent with the student's Lewis structure: 120°

1.A

2.7.A

- (f) SO_2 can be oxidized to form SO_3 according to the following equation. **1 point**



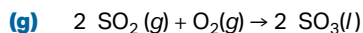
Is the value of ΔS° for the reaction represented above positive or negative? Justify your answer.

One point for indicating negative with a valid explanation that mentions the smaller number of moles of gas in the products.

- Negative, because the reactants are three moles of gas but the products are only two moles of gas.

4.C

9.1.A



1 point

6.D

Is the magnitude of ΔH° for the reaction to form $\text{SO}_3(l)$, represented above, greater than, less than, or equal to the magnitude of ΔH° for the reaction to form $\text{SO}_3(g)$? Justify your answer.

6.9.B

One point for indicating greater magnitude with a valid justification:

- Greater, because the enthalpy of $\text{SO}_3(l)$ is lower than the enthalpy of $\text{SO}_3(g)$ (by an amount equal to the enthalpy of vaporization of $\text{SO}_3(l)$), which makes the difference between the enthalpy of the reactants and the enthalpy of the products a larger amount.

(h) Based on the information above, how does the thermodynamic favorability of the reaction change as the temperature of the reaction system is decreased? Justify your answer.

1 point

5.C

One point for indicating an increased thermodynamic favorability along with a valid justification.

9.3.A

- $\Delta G_{\text{rxn}} = \Delta H_{\text{rxn}} - T\Delta S_{\text{rxn}}$. Assuming that both ΔH_{rxn} and ΔS_{rxn} are constant, as the value of T is decreased, the smaller in value the term $(T\Delta S_{\text{rxn}})$ becomes, making the term $(\Delta H_{\text{rxn}} - T\Delta S_{\text{rxn}})$ more negative. Thus ΔG_{rxn} becomes more negative, increasing the thermodynamic favorability of the reaction.

Total for question 2 10 points